TARIK RAHMAN

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Career Objectives

A Computer Science and Engineering graduate from Dhaka University of Engineering & Technology (DUET), with an MSc research focus on federated learning and deep learning-based medical image segmentation, and prior experience as Adjunct Faculty and Research & Teaching Assistant. I am highly motivated to pursue a career in research and academia, with hands-on experience in university-level teaching, research supervision, and academic publishing. My goal is to contribute to cutting-edge research, solve complex technical challenges, and help shape the future of technology through academic and research endeavors.

EDUCATION

Masters of Science in Computer Science and Engineering 2023 - 2026
Dhaka University of Engineering and Technology (DUET), Gazipur, Bangladesh
Current CGPA: 3.67 out of 4

Bachelor of Science in Computer Science and Engineering 2018 - 2022
Dhaka University of Engineering and Technology (DUET), Gazipur, Bangladesh
CGPA: 3.48 out of 4 (Avg 3.65 in last three semester)

Diploma in Computer Technology 2012-2017
Bogura Polytecnic Institute
CGPA : 3.51 out of 4

Professional Experience

Dhaka University of Engineering & Technology(DUET) Jan 2025 - present
Research Assistant

- Conducting research under the supervision of Dr. Rafiqul Islam on federated learning, deep learning, and medical image segmentation
- Preparing research manuscripts, technical reports, and presentations for academic publication and conference participation
- Conducting experiments, analyzing results, and evaluating model performance using standard segmentation metrics
- Reviewing relevant literature and exploring efficient deep learning and cross-modal learning approaches for medical image analysis

e-Learning and Earning LTD. July 2023 - August 2024
Senior Executive (IT)

- Conducted comprehensive data analysis and developed data pipelines using SQL and Python
- Designed and generated graphs, charts, and dashboards for data visualization using Tableau and Power BI
- Managed IT infrastructure, ensuring high availability, reliability

- Collaborated with cross-functional teams to gather requirements and define project scopes.

Independent University, Bangladesh January 2023 - June 2023
Research and Teaching Assistant

- Conducted research experiments and documented processes
- Analyzed data, summarized results, and generated graphs/spreadsheets
- Fact-checked, proofread manuscripts, and prepared presentations
- Assisted faculty with classroom instruction and laboratory materials
- Graded assignments, provided feedback, and managed records

Independent University, Bangladesh September 2022 - December 2022
Adjunct Faculty

- Conducted four sections of data structure lab classes
- Prepared lesson materials, assignments, and assessments for the students
- Aimed to improve students' performance through effective communication and support

Research Work

(i) **Enhancing Wound Diagnosis with DoubleU-Net: An AI-Driven Segmentation Approach**

(ii) **SLFedUNet: A Super-Lightweight Federated 3D U-Net for Volumetric Brain Tumor Segmentation**

(iii) **An Interpretable CatBoost-Based Model for Accurate Classification of Thyroid Disorders**

(iv) **An Explainable Swin Transformer–Ensemble Framework for Multiclass Alzheimer's Disease Stage Classification from MRI**
Accepted for Presentation at ICCAD 2026 (Lisbon, Portugal)

(v) **A Comprehensive Standardized Dataset on Mental Health Problems (MHPs) of University Students**
Under Review.

(vi) **Undergraduate Research Work**

Thesis Title: "Multiclass Brain Tumor Detection and Improved Classification using Convolutional Neural Networks"

Projects

- **Project of Enterprise Resource Planning (ERP) System:** This is my web development project which has done during my 3rd year 1st semester of undergrad as a sessional course. I have use HTML, CSS, and JS for front end and use PHP for the back end. I have tried to implement some important function and scenario of Enterprise Resource Planning system.
- **Project of Smart Water Heater:** The system makes use of arduino uno board, LCD screen, GSM modem for sending data. This project was running very perfectly.
- **Project of Digital Diary:** I was part of this project. We have used C# and .NET framework to completed this project during 2nd year 2nd semester.

Research Interest

- Artificial Intelligence
- Computer Vision
- Digital Image Processing
- Robotics Perception
- UAV Image/Video Analysis
- Human Computer Interaction

Technical Skills

- **Programming Language:** Python, C, C++
- **Web Programming:** HTML, CSS, PHP
- **Database:** SQL, MySQL
- **Machine Learning:** Classification, Segmentation on 2D and 3D images, Text Classification
- **Deep learning tools**: Tensorflow, Keras, Open CV, pandas, numpy , Matplolib and other related tools
- **Others tools :** Git, Github, SPSS, Latex

Certifications

- Mathematics for Machine Learning: Linear Algebra
- Mathematics for Machine Learning: Multivariate Calculas
- Mathematics for Machine Learning: PCA
- Machine Learning Pipelines with Azure ML Studio

Extra-curricular Activities, Awards and Achievements

- Champion & Selected for the Regional Finals Round at Hult Prize Challenge -2021
- 2nd Runner Up on Robo Rides at DUET CSE Fest-2019
- Hall Management Committee Member, Dr. QK Hall, DUET
- Convenor, Country Tour, Dept. of CSE DUET
- Vice President, Karatoa Students Welfare Association, DUET
- Joint Secretary, Karatoa Students Welfare Association, DUET
- General Secretary, Student Association of Shilkour(SAS)

REFERENCES

Dr. Rafiqul Islam
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DUET,
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Dr. Mohammod Abul Kashem
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